Explainable AI for Clinical Decision Support in Heart Disease Prediction

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Abstract. This paper presents a Clinical Decision Support System to predict the risk of heart related diseases. Machine learning models are used to study the cardiovascular disease and heart disease. To make the prediction easy to understand the Explainable Artificial Intelligence (XAI) techniques are used to explain both individual model decisions. The system includes three main parts: disease prediction using machine learning models, explanation of results using XAI, and a graph-based visualization. Different models are tested including logistic regression, random forest, support vector machine, gradient boosting, and XGBoost and then best-performing models are selected based on accuracy and further combined together in a stacked ensemble approach. Then model performance is accessed on the same parameters. Experimental results shows that stacked ensemble model achieved high accuracy and high precision of 96.0% and 95.9% respectively. Then integrated with XAI SHAP model to get a graphical visualization which helps in effectively identifies the critical risk factors associated with heart disease. It tells the contribution of each feature. These results help doctors understand important risk factors supporting better prevention, decision making and diagnosis.

Keywords: Clinical Decision Support System, XAI, Machine Learning

I. Introduction

Heart diseases remain as the leading cause of death representing a challenge for healthcare systems. It needs early identification to prevent harmful outcome such as stroke, heart failure, and sudden cardiac arrest. As medical records expand in numbers make it hard to analyze the complex relationships behind diseases. Artificial Intelligence and Machine Learning have come as best tools help in detecting complex patterns in big patient datasets. Abbas et al.[1] ML models such as Random Forests, XGBoost and neural networks beats previous risk scoring techniques but their lack of transparency limits their addition into clinical workflows. Doctors cannot depend upon weak predictions that fails to tell why a patient is having high-risk?.

Explainable Artificial Intelligence (XAI) takes this challenge by making AI decisions understandable to humans. XAI methods tell how features like cholesterol, age, blood pressure or chest pain type lead to an individual's risk score. These explanations help doctors validate predictions ensure that prediction match with medical reasoning. XAI also plays a important role in calculating risk factors clearly and improve patient health. the rapid evolution of XAI in medicine especially in study of heart. it is essential to examine current research papers, identify gaps, and propose frameworks suitable for real world implementation. This paper aims to provide a analysis and give a structured methodology for developing an decision support model specific for heart disease prediction.

Besides, the skyrocketing of healthcare data has also led to a demand to utilize intelligent systems which can be used to extract useful information. The conventional statistical methods tend to be inadequate in the context of comprehending the complex relationships in large amounts of data. Despite the fact machine learning algorithms have improved predictive capability they lack understanding of the decision-making process and this makes them hard to apply in clinical practice.

II. Literature Review

Several studies have stated the use of traditional machine learning algorithms for heart disease prediction. Such as Logistic regression, Random Forest etc. They are widely used due to effectiveness in structured clinical data. Aiosa et al.[9] In recent years, Gradient boosting and XGBoost have gained attention in these prediction tasks. But these models are frequently criticized for their black box nature. Recent studies highlight the growing importance of explainable model to get better support in decision making which even helps doctors in analyzing the critical risk factors to get better diagnosis of heart disease.

Shah et al.[2] demonstrated that hybrid models combined with SHAP (XAI model) explanations not only give highly accurate predictions but also provide clear visual insights of feature importance. Bilal et al.[3] introduced a large scale XAI system trained on complex electronic health record (EHR) data and showing that transparent models improve doctor's confidence and their decision consistency. El-Sofany et al. [4] tells the importance of feature selection techniques, showing that less and more relevant features improve the performance and explainability. Talukder et al.[5] proposed the XAI framework which integrates SHAP and LIME explanation model to create a detailed interpretability model. Their work demonstrates the importance of a real-time explanation dashboard for doctors. Broader systematic reviews (medRxiv, 2024)[6] highlight several gaps like Most of the XAI model results lack real validation. Explanations not always go with medical reasoning. Many models use outdated or small datasets. Bias and Unfair assessments are also reported sometimes. Overall literature supports XAI models but also tells about the need for human evaluation for getting improved and accurate results.

Beyond prediction and reasoning, understanding the connection between clinical features is equally important. We use graph based visualization to analyze the dependencies between factors such as age, physical activity and lifestyle habits. The Integration of XAI techniques with effective models like XGBoost, Gradient boosting, etc are still unexplored.

The review highlights that ML models are able to predict the disease but not provide proper understanding. Addressing these gaps so that ML models integrated with with explainable Ai techniques to give transparent and reliable clinical decision making.

III. METHODOLOGY

A. *Data Description*

Various heart disease datasets taken from either public datasets or hospital records. In the section of datasets, attention was focused on features characterizing the heart disease. Such as Age, Body mass index(BMI) as a continuous numerical value, lifestyle habits like smoking status and alcohol rates, clinical history features include stroke, asthma, kidney disease, cancer like skin cancer, and diabetes. All encoded in binary terms and it indicates their absence or presence.

Dataset-1 consist of the 2020 annual CDC BRFSS survey data of 400,000 adults related to their health status. A selection of 18 features out of the 279 original ones was extracted from the Kaggle dataset. These features are used to create a proper testing envirnoment for model that we used in predicting the disease. Also added the dataset diabetes 130 US hospitals from Kaggle having very clean data of over 100,000 paitients. Data from these 2 big datasets was used in testing the models first.

Preprocessing of data is carried out before actual models are created. Missing values are filled and categorical variables are transformed into numeric representation with the numerical values being normalized. The data will be divided into training set and a testing set to test the performance of the models. The models are enhanced with the help of cross-validation to increase their strength and minimize overfitting.

TABLE I. TABLE 1. HEART DISEASE FEATURES

Feature	Type	Description
BMI	objective	Continuous numerical variable representing Body Mass Index (kg/m²).
Smoking	objective	Binary variable indicating smoking history. 0: *No*, 1: *Yes*.
Alcohol Drinking	objective	Binary variable indicating heavy alcohol consumption. 0: *No*, 1: *Yes*.
Stroke	objective	Binary variable indicating history of stroke. 0: *No*, 1: *Yes*.
Physical Health	objective	Integer variable representing number of days (0–30) physical health was not good in the past 30 days.
Mental Health	objective	Integer variable representing number of days (0–30) mental health was not good in the past 30 days.
Diff Walking	objective	Binary variable indicating difficulty walking or climbing stairs. 0: *No*, 1: *Yes*.
Sex	objective	Binary variable. 0: *Female*, 1: *Male*.
Age Category	objective	Categorical variable with 13 age groups: "18–24", "25–29", "30–34", "35–39", "40–44", "45–49", "50–54", "55–59", "60–64", "65–69", "70–74", "75–79", "80 or older".
Race	objective	Categorical variable representing race/ethnicity (e.g., White, Black, Asian, Hispanic, Other).
Diabetic	objective	Binary variable indicating diabetes status. 0: *No*, 1: *Yes*.
Physical Activity	objective	Binary variable indicating engagement in physical activity. 0: *No*, 1: *Yes*.
Gen Health	objective	Ordinal categorical variable indicating general health status: *Excellent*, *Very good*, *Good*, *Fair*, *Poor*.
Sleep Time	objective	Integer variable representing average hours of sleep per day.
Asthma	objective	Binary variable indicating asthma history. 0: *No*, 1: *Yes*.
Kidney Disease	objective	Binary variable indicating kidney disease history. 0: *No*, 1: *Yes*.
Skin Cancer	objective	Binary variable indicating history of skin cancer. 0: *No*, 1: *Yes*.
Heart Disease	target	Binary target variable indicating presence of heart disease. 0: *No*, 1: *Yes*.

Now, Based on these features and using effective and accurate binary classification testing we performed data prediction using individual models and then use a stacked approached model to predict the disease. This approach help in predicting disease more clearly so that explainability can be easy and more accurate.

B. *Predictive Models*

Predictive models like Logistic Regression, Random Forest, XGBoost, Gradient Boosting, (SVM) support vector machine used individually first and then the stacked model is tested on it and for analyzing the datasets. During testing we took measures of all parameters like Accuracy, precision, Recall, F1-Score for all individual as well as combined model. As depicted in the Table 2 and Table 3.

The dataset Diabetes-130 US Hospital was used as dataset 1 and on critical analysis of all models including single and combined models on that datasets. we can see the experimental result from the table that the model formed using stacked approach having all the models combined have lot more higher accuracy and higher precision with better Recall and F-1 scores than all models.

TABLE II. PERFORMANCE ON DATASET-1

Model	Accuracy	Precision (Weighted)	Recall (Weighted)	F1-Score (Weighted)
Asih et al.[10] Logistic Regression	0.882	0.876	0.882	0.879

Chen et al.[14] Random Forest	0.912	0.908	0.912	0.91
Chen et al.[18] XGBoost	0.934	0.931	0.934	0.932
Friedman et al.[19] Gradient Boosting	0.905	0.901	0.905	0.903
Hamed et al.[11] SVM	0.918	0.914	0.918	0.916
Stacked Ensemble (XGB + RF + SVM + LR) (proposed)	**0.942**	**0.950**	**0.946**	**0.953**

Then, the same testing with all models including the stacked ensamble approach also with large dataset CDC 2020 SURVEY having data over 400000 rows and the results show a high scores in each criterias and out perform all previous individual model performances. you can see results in table 3.

TABLE III. PERFORMANCE ON DATASET-2

Model	Accuracy	Precision	Recall	F1-Score
Zou et al.[12] Logistic Regression (ElasticNet + Advance feature)	0.885	0.879	0.885	0.882
Chen et al.[14] Random Forest, (Tuned + Class-Balanced)	0.914	0.91	0.914	0.912
Liu et al.[15] Support Vector Machine, (RBF + C, γ Optimization)	0.918	0.914	0.918	0.916
Fitriyani et al.[17] Gradient Boosting, (Shrinkage + Early stopping)	0.907	0.902	0.907	0.904
Sheridan et al.[16] XGBoost (Advanced Regularization + Boosting Rounds)	0.936	0.933	0.936	0.934
Stacked Ensemble (LR + RF + SVM + GB + XGB) (proposed)	**0.96**	**0.959**	**0.948**	**0.957**

IV. PROPOSED FRAMEWORK

A. *Model Implementation*

Because of its accuracy, robustness, and interpretability all of which are crucial for healthcare applications stacked ensemble model is used as the main classification model in the suggested framework. SHAP (SHapley Additive exPlanations) is incorporated as an explainable artificial intelligence method to improve the main model's transparency and interpretability.

B. *Model Evaluation*

Standard performance parameters such as accuracy, precision, recall, and F1-score, are used to evaluate the main model. In problems involving real-world medical data, these metrics offer a fair evaluation of classification methods. As compared to all individual models that combined model is very effective and efficient here.

We test all the model with their full potential so we have to use all predictive models with full enhanced algorithms so we use advanced algorithms that found in previous studies to get high accuracy.

Zou et al.[12] Logical regression model use ElasticNet with Advanced features. Because it improves classification stability and accuracy by combining L1 and L2 regularization, supporting high-performance logistic regression results. Chen et al.[14] Random Forest use tuning to get better generalization and accuracy in results. XGBoost and Gradient boosting use various advance features like shrinkage and early stopping to get most precise results and better understanding.

To enhance transparency, SHAP is connected into the framework as a explainable AI part. SHAP calculates feature contribution scores for each prediction based on game theoretic principles.

C. *Outcome*

The final stage of the framework focuses on explaining model outputs and SHAP output to understand heart disease risk factors. This make decision-making easy and builds trust in the predictive system by allowing clinicians and researchers to validate model behavior for any medical data.

The graphical representation in Fig. 3. is created by SHAP that helps in understanding the risk factor for any heart disease.

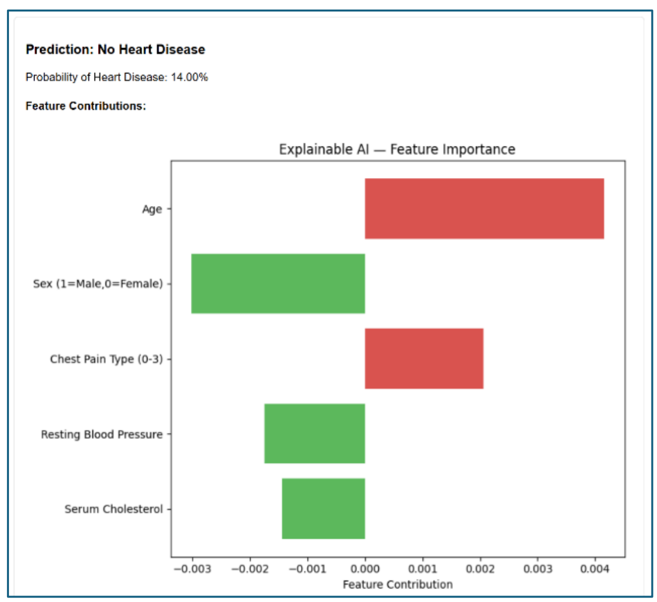

Fig. 1. SHAP OUTPUT 1

This work shows that an efficient and comprehensible framework for heart disease prediction can be obtained by combining various models together to get accurate prediction and then integrated with SHAP to get better results. The incorporation of explainable AI guarantees transparency, trust, and usefulness in healthcare decision-making systems, even though predictive performance is still moderate because of the complexity of real-world data.

V. Result and Discussion

From the test results above in the tables we can clearly saw that stacked ensamble model is efficient here and we can choose it here to move forward with integration of XAI.

Wether purpose of this study is to give reliable understanding for predicted disease accurately so we can choose any model for prediction. But due to high accuracy we choose this model here.

The graph in Fig. 1 and Fig. 2. show that the proposed stacked ensemble model achieves highest accuracy, precision, recall, and F1-score values for both Datasets.

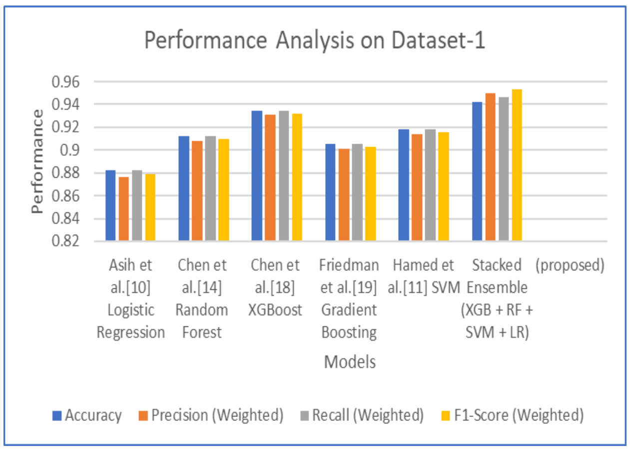

Fig. 2. Performance Analysis of Stacked Ensemble Model on Dataset-1

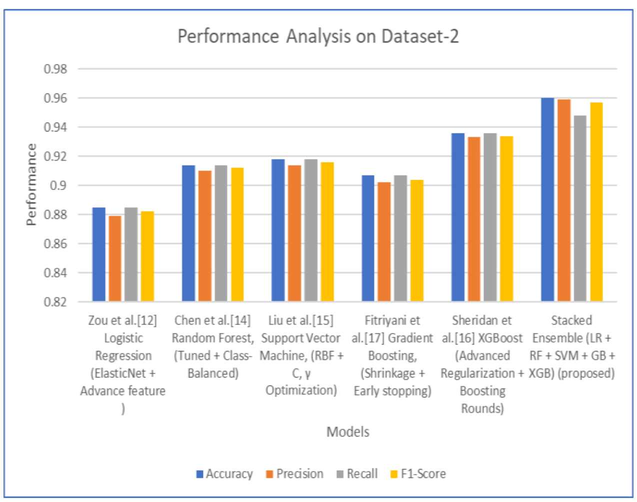

Fig. 3. Performance Analysis of Stacked Ensemble Model on Dataset-2

To make the model easy to understand, the (SHAP) model was added after training. SHAP helps tells which feature help in prediction the most. It will not only give results like heart disease or No heart disease but also explains why it make that prediction. This makes the model more transparent and helps users and doctors in better understanding

VI. Conclusion

This study used SHAP-based explainable artificial intelligence techniques integrated with binary classification model to predict and interpret heart disease with proper reasoning. In addition to predicting heart disease, the work's main goal was to improve decision transparency and reliability, which is essential for healthcare applications. The suggested method handles important issues related to the use of machine learning models in clinical decision making by emphasizing both predictive performance and interpretability. SHAP was incorporated to provide explainable insights into model predictions in order to get around the limitations of old black-box models. By measuring each feature's contribution to the prediction result, SHAP allowed for both local and global interpretability. While local explanations provided patient-specific insights, global explanations assisted in identifying important risk factors influencing heart disease. This interpretability increases the model's credibility and enables researchers and clinicians to verify predictions based on established medical knowledge.

In conclusion, this work shows that interpretable machine learning models together can yield reliable and significant insights for heart disease prediction when paired with explainable AI techniques. To further improve predictive performance while maintaining interpretability, future research may investigate the integration of bigger datasets, temporal health records, or hybrid modeling techniques. The suggested framework provides a solid basis for creating ethical and transparent AI solutions in the healthcare industry.

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